David Arnaiz

Embedded Software Engineer

Madrid, Spain (relocating soon) | electronplant@gmail.com
ElectronPlant | david-arnaiz-martínez | Personal Website

Skills

- Programming Languages
  - Proficient: C.
  - Fluent: C++, Python, Rust.
  - Interm: LaTeX, Assembly, shell.
  - Basic: VHDL.
- Serial protocols (I2C, SPI, UART)
- Development Tools:
  - IDEs: VS Code, Eclipse.
  - EDA: KiCad, Cadence.
  - CI/CD: Git
  - Dev: VS Code Copilot
- Debugging Tools:
  - Oscilloscopes.
  - Logic analyzers.
  - Power monitors.
  - GDB, SWD.
- Languages:
  - Spanish: Native.
  - English Professional Proficiency (daily corporate use, written publications).

Achievements

- Best academic record award (2018).
- Industrial Doctorate program (2019 DI 075).
- HP R&D Academy program

Publications

[1] Arnaiz et al. (2021). Data Relevance-Aware Dynamic Sensing Technique with Battery Lifetime Guarantee for Wireless Sensor Nodes. DCIS2021. IEEE.
[2] Arnaiz et al. (2023). Relating Context- and Self- Awareness in the Internet of Things. CoopIS (pp. 384-402).Springer.
[3] Arnaiz et al. (2023). On Anomaly-Aware Structural Health Monitoring at the Extreme Edge. IEEE Access.
[4] Arnaiz et al. (2024). Energy and relevance–aware adaptive monitoring method for wireless sensor nodes with hard energy constraints. Integration. The VLSI journal, 94(article 102097).

Profile

- Embedded system engineer with 8+ years of experience.
- Ph.D. in self-aware wireless sensor nodes.
- Proficient in C; with C++, Rust, Python, LaTeX and assembly experience.
- Proven track record with:
  - Defining embedded software architectures.
  - C drivers and Real-time FreeRTOS applications.
  - Mixed-signal electronic schematic design.
  - Generating agentic workflows to automate routine tasks.
- Team player, working as part of international and multidisciplinary teams.
- Great communication, coordination and interpersonal skills.
- Focused on continuous improvement and innovation, delivering contributions to meaningful projects.

Work Experience

Embedded Firmware Engineer [Dec. 2019 - Present]
Worldsensing Barcelona, Spain

- Defined and developed the FW architecture for the vibration node featuring a configurable data processing pipeline, and integrating different firmware modules and RTOS tasks.
- Developed multiple sensor drivers, fully unit tested FW modules, and complete features for low-power, LoRa-based wireless sensor nodes.
- Designed and implemented event-driven sensing feature for the complete wireless sensor family.

Electronic Designer [Oct. 2018 - Dec. 2019]
Rolls-Royce - Control Systems (via Alten) Birmingham, UK

- Designed lightning strike protection circuits for high-reliability aircraft electronics within the Engine Monitoring Unit (EMU).
- Authored hardware design documents and guidelines for safety-critical systems.
- Conducted hands-on laboratory testing of electronic prototypes.

R&D Electronic Engineer [Sept. 2016 - Sept. 2018]
HP - Large Format Printing Barcelona, Spain

- Owned the electronic schematic design for printer mixed-signal, mechatronics PCBs based on STM32 MCUs.
- Developed custom HW and FW prototypes for inductive and optical sensing.
- Defined hardware diagnostic guidelines and printer self-diagnostics schemes.

Education

Ph.D. in Electronic Engineering [Sept. 2019 - July. 2024]
Polytechnic University of Cataluña (UPC) Barcelona, Spain

- Grade of excellence.
- Thesis in dynamic energy management using self-awareness in wireless sensor nodes.

M.Sc. in Electronic Engineering [Sept. 2016 - Sept. 2018]
Polytechnic University of Cataluña (UPC) Barcelona, Spain

- GPA of 3.752 (out of 4).

B.Sc. in Electronic Engineering [Sept. 2012 - Sept. 2016]
University of Granada (UGR) Granada, Spain

- GPA of 3.491 (out of 4).